

There are a total of six problems. You have to solve all of them.

Problem 1 (CO5): Pumping Lemma (5 points)

Let  $\Sigma = \{0, 1\}$ . Consider the following language.

$$L = \{w \in \Sigma^* : w = 1^a 0^b 0^c 1^d, \text{ where } a + d = b + c \text{ and } a, b, c, d \geq 0\}$$

Use the pumping lemma to **demonstrate** that  $L$  is not regular language.

Problem 2 (CO3): Designing Context-Free Grammars (10 points)

Let  $\Sigma = \{0, 1\}$ . Consider the following languages. Recall that for a string  $w$ ,  $|w|$  denotes the length of  $w$ .

$$L_1 = \{w \in \Sigma^* : w \text{ is a odd length palindrome}\}$$

$$L_2 = \{w \in \Sigma^* : \text{length of } w \text{ is multiple of two}\}$$

$$L_3 = \{x000y : x, y \in L_2, |x| = |y|\}$$

$$L_4 = L_1 \cap L_3$$

Now solve the following problems.

- (a) **Give** a context-free grammar for the language  $L_1$ . (3 points)
- (b) **Give** a context-free grammar for the language  $L_3$ . (4 points)
- (c) **Give** a context-free grammar for the language  $L_4$ . (3 point)

Problem 3 (CO4): The CYK Algorithm (5 points)

**Apply** the CYK algorithm to fill up the table for the string aabaa using the following grammar. Here a and b are terminals, and the rest are variables.

$$\begin{aligned} S &\rightarrow BA \mid BC \\ A &\rightarrow AB \mid AC \mid a \\ B &\rightarrow CB \mid CC \mid b \\ C &\rightarrow CA \mid a \end{aligned}$$

|                    |                  |                |                    |                  |
|--------------------|------------------|----------------|--------------------|------------------|
| 1,5<br>$\{?\}$     |                  |                |                    |                  |
| 1,4<br>$\{?\}$     | 2,5<br>$\{S,A\}$ |                |                    |                  |
| 1,3<br>$\{?\}$     | 2,4<br>$\{S,A\}$ | 3,5<br>$\{?\}$ |                    |                  |
| 1,2<br>$\{A,B,C\}$ | 2,3<br>$\{?\}$   | 3,4<br>$\{S\}$ | 4,5<br>$\{A,B,C\}$ |                  |
| 1,1<br>$\{A,C\}$   | 2,2<br>$\{A,C\}$ | 3,3<br>$\{B\}$ | 4,4<br>$\{A,C\}$   | 5,5<br>$\{A,C\}$ |
| $a$                | $a$              | $b$            | $a$                | $a$              |

Problem 4 (CO3): Constructing Pushdown Automata (10 points)

Let  $\Sigma = \{0, 1\}$ . Consider the following languages.

$$L_1 = \{w \mid w \text{ starts and ends with the different character}\}$$

$$L_2 = \{w \mid \text{the number of 0s in } w \text{ is not same as the number of 1s}\}$$

- (a) **Give** the state diagram of a pushdown automaton that recognizes  $L_1$ . (4 points)
- (b) **Give** the state diagram of a pushdown automaton that recognizes  $L_2$ . (6 points)

Problem 5 (CO3): Derivations, Parse Trees and Ambiguity (10 points)

Take a look at the grammar below and solve the following problems.

$$A \rightarrow 1A \mid 0B \mid 00A$$

$$B \rightarrow 1B \mid 1C \mid 0A \mid 00B$$

$$C \rightarrow 0C0 \mid 0C1 \mid 1C0 \mid 1C1 \mid 0 \mid 1$$

- (a) **Give** a leftmost derivation for the string 10110010. (3 points)
- (b) **Sketch** the parse tree corresponding to the derivation you gave in (a). (2 points)
- (c) **Demonstrate** that the given grammar is ambiguous by showing two more parse trees (apart from the one you already found in (b)) for the same string. (3 points)
- (d) **Find** a string  $w$  of length six such that  $w$  has exactly one parse tree in the grammar above. (1 point)
- (e) **Design** an unambiguous Context Free Grammar for the language represented by the given ambiguous grammar. (1 point)

Problem 6 (CO4): Chomsky Normal Form (10 points)

Answer the following questions.

- (a) **List** the productions that violate the conditions of the Chomsky Normal Form (CNF) in the following grammar. (5 points)

$$P \rightarrow a \mid PR \mid \varepsilon$$

$$Q \rightarrow bP \mid R$$

$$R \rightarrow QQ \mid aT$$

$$T \rightarrow QT \mid \varepsilon$$

- (b) **Write** down the additional rules that need to be added to the following grammar if the production  $A \rightarrow \varepsilon$  is removed. (3 points)

$$S \rightarrow AA \mid bB$$

$$A \rightarrow CB \mid aA \mid \varepsilon$$

$$B \rightarrow ABA \mid b \mid \varepsilon$$

- (c) **Write** down the additional rules that need to be added to the following grammar if all the unit productions are removed. (2 points)

$$A \rightarrow 0A1B1 \mid C$$

$$B \rightarrow 1B \mid B$$

$$C \rightarrow BC \mid A \mid B$$